

PERIODIC TABLE OF CYBERSECURITY



The “Periodic Table of Cybersecurity” organizes the entire security ecosystem into distinct categories: **Threats, Vulnerabilities, Tools, Frameworks & Standards, Concepts & Operations, Defensive Measures, and Roles & Protocols.**

This report explains **each element**, its purpose, and how it contributes to cybersecurity.

1. THREATS (Red Elements)

These represent malicious activities or attack vectors used by threat actors.

Code	Threat	Explanation
PH	Phishing	Social engineering attack using emails/messages to steal credentials or data.
MW	Malware	Any malicious software (worms, trojans, RATs) designed to harm systems.
RW	Ransomware	Malware that encrypts data and demands ransom for decryption keys.
CS	Cross-site Scripting (XSS)	Injection attack where malicious scripts run in a victim’s browser.
DD	DDoS	Distributed Denial of Service—flooding a server to take it offline.
SW	Spyware	Malware that covertly collects user data or activity.
AD	Adware	Unwanted advertising software; often intrusive or malicious.
TN	Trojan	Malware disguised as legitimate software to execute harmful actions.
RK	Rootkit	Stealthy malware hiding processes/files; used for persistence.
KL	Keylogger	Captures keystrokes to steal credentials.

2. VULNERABILITIES (Orange Elements)

Weaknesses in systems that attackers exploit.

Code	Vulnerability	Explanation
US	Unpatched Software	Missing security updates that expose systems.
OP	Open Ports	Unsecured ports accessible from the network.
BA	Broken Authentication	Weak or flawed authentication mechanisms.
PE	Privilege Escalation	Gaining higher-level permissions illegally.
CJ	Clickjacking	UI redress attack tricking users into unintended actions.
NM	Network Misconfigurations	Faulty settings enabling unauthorized access.
BS	Broken Sessions	Poor session management, enabling hijacking.
JR	Insecure APIs (John the Ripper for cracking context)	APIs lacking proper security controls.
MZ	Misconfigurations	General system/app/server misconfigurations.
ST	Sensitive Data Exposure	Poor protection of critical data (encryption missing).
MF	Misconfigured Firewall	Weak rules allow attackers to penetrate.
DE	Default Credentials	Default usernames/passwords still in use.
XS	XSRF/CSRF	Attack forcing users to execute unwanted actions.
HR	Hardcoded Credentials	Embedded passwords in code or binaries.
RC	Remote Code Execution	Ability to run arbitrary commands on systems.
WS	Weak Security Controls	Policies or configurations insufficient for protection.
HD	Hidden Directories	Unprotected directories exposing sensitive data.
AG	Algorithm Weakness	Outdated or weak cryptographic algorithms.
TD	TDoS	Telephony Denial of Service—overloading VoIP or phone systems.

Code	Vulnerability	Explanation
ID	Insecure Data Flow	Poor handling of data transfer enabling leaks.
IA	Insecure APIs	Lack of authentication, validation, or encryption.
WT	Weak Transport Security	Outdated protocols like SSL 3.0.
BO	Buffer Overflow	Memory attack leading to crashes or code execution.
XM	XML External Entity Attack	XXE vulnerabilities in XML parsers.
IM	Insufficient Monitoring	Lack of visibility into network/system activity.

3. TOOLS (Green Elements)

Common cybersecurity tools for testing, analysis, and exploitation.

Code	Tool	Explanation
NM	Nmap	Network scanning and reconnaissance tool.
BS	Burp Suite	Web vulnerability scanning and interception framework.
JR	John the Ripper	Password cracking tool.
MZ	Metasploit	Penetration testing and exploitation framework.
ST	Snort	Network intrusion detection/prevention system (IDS/IPS).
NM	Nikto	Web server vulnerability scanner.
HC	Hashcat	GPU-accelerated password cracker.
SQ	SQLMap	Automated SQL injection exploitation tool.
KX	Kali Linux	Linux distro for penetration testing.
MS	Metasploit (duplicate category)	Used for exploitation and payload delivery.
HD	Hydra	Password brute-force tool supporting multiple protocols.
AG	Aircrack-ng	Wireless network security auditing tool.

Code	Tool	Explanation
GR	Gobuster	Directory, file, and DNS brute-forcing tool.
TD	Tcpdump	Network packet capture and analysis tool.

4. FRAMEWORKS & STANDARDS (Blue Elements)

Security guidelines, best practices, and compliance frameworks.

Code	Framework	Explanation
ME	MITRE ATT&CK	Matrix of adversary tactics, techniques, and procedures.
IS	ISO/IEC 27001	Global standard for information security management.
OW	OWASP Top 10	Top 10 critical web security risks.
CT	COBIT	Framework for governance and management of IT.
PS	PCI DSS	Payment Card Industry Data Security Standard.
NC	NIST CSF	Cybersecurity framework for risk management.
S2	SOC 2	Compliance for service providers handling customer data.
HP	HIPAA	Health information privacy and security regulation.
GD	GDPR	European data protection regulation.
ZT	Zero Trust Model	"Never trust, always verify" security architecture.
CY	CIS Controls	Best-practice security controls for organizations.
AR	Audit & Reporting	Compliance and auditing methodology.

5. CONCEPTS & OPERATIONS (Grey Elements)

Core principles and operational methodologies in cybersecurity.

Code	Concept	Explanation
SB	Sandboxing	Isolating apps/files to prevent system compromise.
EP	Encryption	Converting data into unreadable format for protection.
FL	Firewall	Filters traffic to enforce security rules.
TI	Threat Intelligence	Analyzing threat data to predict/prevent attacks.
VA	Vulnerability Assessment	Systematic identification of weaknesses.
SE	Security Monitoring	Continuous monitoring for threats.
VP	VPN	Secure communication tunnel over public networks.
Ed	Endpoint Detection (EDR)	Detects and responds to endpoint threats.
SM	SIEM	Security Information and Event Management.
PM	Patch Management	Keeping systems updated against vulnerabilities.
IR	Incident Response	Handling and mitigating security incidents.
MA	Multi-Factor Authentication	Extra layers of login security.
DL	Data Loss Prevention	Prevents unauthorized data access or exfiltration.

6. ROLES & PROTOCOLS (Black Elements)

Cybersecurity job roles and technical protocols.

Code	Role	Explanation
PT	Penetration Tester	Performs ethical hacking to find vulnerabilities.
SA	Security Analyst	Monitors and protects systems from threats.
SO	SOC Analyst	Operates in SOC to detect and respond to alerts.
IM	IAM Specialist	Manages identity and access.

Code	Role	Explanation
RT	Red Teamer	Simulates real-world attacks for defense improvement.
BT	Blue Teamer	Defends systems against attacks.
NT	Network Engineer	Manages secure network infrastructure.
SE	SIEM Engineer	Designs and maintains SIEM systems.
GM	GRC Analyst	Governance, Risk, and Compliance specialist.
FL	Forensic Investigator	Investigates digital crimes.
DO	DevSecOps	Integrates security in CI/CD pipelines.
NE	Network Security Engineer	Protects network layers and protocols.
FI	Forensic Investigator	Collects/analyzes digital evidence.
SC	Cloud Security	Protects cloud environments and workloads.
CL	CISO	Chief Information Security Officer.

7. Conclusion

The *Periodic Table of Cybersecurity* provides a powerful overview of the entire security ecosystem.

Understanding each element helps:

- ✓ Strengthen security posture
- ✓ Identify vulnerabilities early
- ✓ Build effective incident response
- ✓ Improve compliance and governance
- ✓ Develop a structured cybersecurity learning path

This table acts as a **roadmap for professionals, learners, and organizations** to navigate the complex world of cybersecurity.